

1. The greatest integer which divides the number $101^{100} - 1$, 9. If the coefficients of T_r, T_{r+1}, T_{r+2} terms of $(1+x)^{14}$ are in

FOR:

AIEEE/PET/PMT/ PAT/NDA/AFMC/ PPT

is

A.P., then $r =$

[Pb. CET 2002]

Binomial

SIGMA Theorem BIAORA

- (a) 100 (b) 1000 [MP PET 1998] (a) 6 (b) 7
(c) 10000 (d) 100000 (c) 8 (d) 9
2. The positive integer just greater than $(1 + 0.0001)^{10000}$ is [AIEEE 2002]
- (a) 4 (b) 5
(c) 2 (d) 3
3. The last digit in 7^{300} is [Karnataka CET 2004]
- (a) 7 (b) 9
(c) 1 (d) 3
4. If the ratio of the coefficient of third and fourth term in the expansion of $\left(x - \frac{1}{2x}\right)^n$ is 1 : 2, then the value of n will be
- (a) 18 (b) 16
(c) 12 (d) -10
5. If the coefficients of r^{th} term and $(r+4)^{th}$ term are equal in the expansion of $(1+x)^{20}$, then the value of r will be [RPET 1985, 97; Kerala (Engg.) 2001; MP PET 2002]
- (a) 7 (b) 8
(c) 9 (d) 10
6. If coefficient of $(2r+3)^{th}$ and $(r-1)^{th}$ terms in the expansion of $(1+x)^{15}$ are equal, then value of r is [RPET 1995, 2003; UPSEAT 2001]
- (a) 5 (b) 6
(c) 4 (d) 3
7. If x^4 occurs in the r^{th} term in the expansion of $\left(x^4 + \frac{1}{x^3}\right)^{15}$, then $r =$ [MP PET 1995; Pb. CET 2002]
- (a) 7 (b) 8
(c) 9 (d) 10
8. The first 3 terms in the expansion of $(1+ax)^n$ ($n \neq 0$) are 1, $6x$ and $16x^2$. Then the value of a and n are respectively [Kerala (Engg.) 2002]
- (a) 2 and 9 (b) 3 and 2
(c) $2/3$ and 9 (d) $3/2$ and 6
10. If the coefficients of p^{th} , $(p+1)^{th}$ and $(p+2)^{th}$ terms in the expansion of $(1+x)^n$ are in A.P., then [AIEEE 2005]
- (a) $n^2 - 2np + 4p^2 = 0$
(b) $n^2 - n(4p+1) + 4p^2 - 2 = 0$
(c) $n^2 - n(4p+1) + 4p^2 = 0$
(d) None of these
11. The ratio of the coefficient of terms $x^{n-r}a^r$ and $x^r a^{n-r}$ in the binomial expansion of $(x+a)^n$ will be
- (a) $x : a$ (b) $n : r$
(c) $x : n$ (d) None of these
12. If p and q be positive, then the coefficients of x^p and x^q in the expansion of $(1+x)^{p+q}$ will be [MNR 1983; AIEEE 2002]
- (a) Equal
(b) Equal in magnitude but opposite in sign
(c) Reciprocal to each other
(d) None of these
13. The coefficient of x^{-7} in the expansion of $\left(ax - \frac{1}{bx^2}\right)^{11}$ will be [IIT 1967; RPET 1996; Pb. CET 2003]
- (a) $\frac{462a^6}{b^5}$ (b) $\frac{462a^5}{b^6}$
(c) $\frac{-462a^5}{b^6}$ (d) $\frac{-462a^6}{b^5}$
14. If the coefficients of x^7 and x^8 in $\left(2 + \frac{x}{3}\right)^n$ are equal, then n is [EAMCET 1983; Kurukshetra CEE 1998; DCE 2000; RPET 2001; UPSEAT 2001]
- (a) 56 (b) 55
(c) 45 (d) 15
15. If in the expansion of $(1+x)^m(1-x)^n$, the coefficient of x and x^2 are 3 and -6 respectively, then m is

[IIT 1999; MP PET 2000]

- (a) 6 (b) 9
(c) 12 (d) 24

16. If x^m occurs in the expansion of $\left(x + \frac{1}{x^2}\right)^{2n}$, then the

coefficient of x^m is

[UPSEAT 1999]

- (a) $\frac{(2n)!}{(m)!(2n-m)!}$ (b) $\frac{(2n)!3!3!}{(2n-m)!}$
(c) $\frac{(2n)!}{\left(\frac{2n-m}{3}\right)!\left(\frac{4n+m}{3}\right)!}$ (d) None of these

17. In the expansion of $(1+x+x^3+x^4)^{10}$, the coefficient of x^4 is

[MP PET 2000]

- (a) $^{40}C_4$ (b) $^{10}C_4$
(c) 210 (d) 310

18. If the coefficient of 4th term in the expansion of $(a+b)^n$ is 56, then n is

[AMU 2000]

- (a) 12 (b) 10
(c) 8 (d) 6

19. The term independent of x in the expansion of

$$\left(\sqrt{\frac{x}{3}} + \frac{3}{2x^2}\right)^{10} \text{ will be}$$

[IIT 1965; BIT Ranchi 1993; KCET 2000; UPSEAT 2001]

- (a) $3/2$ (b) $5/4$
(c) $5/2$ (d) None of these

20. If the middle term in the expansion of $\left(x^2 + \frac{1}{x}\right)^n$ is $924x^6$,

then $n =$

- (a) 10 (b) 12
(c) 14 (d) None of these

21. The middle term in the expansion of $(1+x)^{2n}$ is

[DCE 2002]

- (a) $\frac{(2n)!}{n!}x^2$ (b) $\frac{(2n)!}{n!(n-1)!}x^{n+1}$
(c) $\frac{(2n)!}{(n!)^2}x^n$ (d) $\frac{(2n)!}{(n+1)!(n-1)!}x^n$

22. The greatest coefficient in the expansion of $(1+x)^{2n+2}$ is

[BIT Ranchi 1992]

- (a) $\frac{(2n)!}{(n!)^2}$ (b) $\frac{(2n+2)!}{\{(n+1)!\}^2}$
(c) $\frac{(2n+2)!}{n!(n+1)!}$ (d) $\frac{(2n)!}{n!(n+1)!}$

23. The coefficient of t^{24} in the expansion of $(1+t^2)^{12}(1+t^{12})(1+t^{24})$ is

[IIT Screening 2003]

- (a) $^{12}C_6 + 2$ (b) $^{12}C_5$
(c) $^{12}C_6$ (d) $^{12}C_7$

24. The coefficient of x^n in expansion of $(1+x)(1-x)^n$ is

[AIEEE 2004]

- (a) $(-1)^{n-1}n$ (b) $(-1)^n(1-n)$
(c) $(-1)^{n-1}(n-1)^2$ (d) $(n-1)$

25. The middle term in the expansion of $\left(x + \frac{1}{2x}\right)^{2n}$, is

[MP PET 1995]

- (a) $\frac{1.3.5....(2n-3)}{n!}$ (b) $\frac{1.3.5....(2n-1)}{n!}$
(c) $\frac{1.3.5....(2n+1)}{n!}$ (d) None of these

26. The middle term in the expression of $\left(x - \frac{1}{x}\right)^{18}$ is

[Karnataka CET 2005]

- (a) $^{18}C_9$ (b) $^{-18}C_9$
(c) $^{18}C_0$ (d) $^{-18}C_{10}$

27. $^{10}C_1 + ^{10}C_3 + ^{10}C_5 + ^{10}C_7 + ^{10}C_9 =$

[MP PET 1982]

- (a) 2^9 (b) 2^{10}
(c) $2^{10} - 1$ (d) None

28. of these $^nC_0 - \frac{1}{2}^nC_1 + \frac{1}{3}^nC_2 - \dots + (-1)^n \frac{^nC_n}{n+1} =$

- (a) n (b) $1/n$
(c) $\frac{1}{n+1}$ (d) $\frac{1}{n-1}$

29. If $(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$, then $C_0^2 + C_1^2 + C_2^2 + C_3^2 + \dots + C_n^2 =$

[MP PET 1985; Karnataka CET 1995; MNR 1999]

- (a) $\frac{n!}{n!n!}$ (b) $\frac{(2n)!}{n!n!}$
(c) $\frac{(2n)!}{n!}$ (d) None of these

30. $C_1 + 2C_2 + 3C_3 + 4C_4 + \dots + nC_n =$

[RPET 1995; MP PET 2002; Orissa JEE 2005]

- (a) 2^n (b) $n \cdot 2^n$
(c) $n \cdot 2^{n-1}$ (d) $n \cdot 2^{n+1}$

31. $\frac{C_0}{1} + \frac{C_1}{2} + \frac{C_2}{3} + \dots + \frac{C_n}{n+1} =$

[RPET 1996]

- (a) $\frac{2^n}{n+1}$ (b) $\frac{2^n - 1}{n+1}$
(c) $\frac{2^{n+1} - 1}{n+1}$ (d) None of these

32. $\frac{1}{1!(n-1)!} + \frac{1}{3!(n-3)!} + \frac{1}{5!(n-5)!} + \dots =$

[AMU 2005]

- (a) $\frac{2^n}{n!}$; for all even values of n
(b) $\frac{2^{n-1}}{n!}$; for all values of n i.e., all even odd values
(c) 0
(d) None of these

33. If $(1+x)^{15} = C_0 + C_1x + C_2x^2 + \dots + C_{15}x^{15}$, then $C_2 + 2C_3 + 3C_4 + \dots + 14C_{15} =$

[IIT 1966]

- (a) $14 \cdot 2^{14}$ (b) $13 \cdot 2^{14} + 1$
(c) $13 \cdot 2^{14} - 1$ (d) None of these

34. The value of $\frac{C_1}{2} + \frac{C_3}{4} + \frac{C_5}{6} + \dots$ is equal to

[Karnataka CET 2000]

- (a) $\frac{2^n - 1}{n+1}$ (b) $n \cdot 2^n$
(c) $\frac{2^n}{n}$ (d) $\frac{2^n + 1}{n+1}$

35. The sum of all the coefficients in the binomial expansion of $(x^2 + x - 3)^{319}$ is

[Bihar CEE 1994]

- (a) 1 (b) 2
(c) -1 (d) 0

36. If the sum of the coefficients in the expansion of $(x - 2y + 3z)^n$ is 128 then the greatest coefficient in the expansion of $(1+x)^n$ is

- (a) 35 (b) 20
(c) 10 (d) None of these

37. If n is an integer greater than 1, then

$$a^{-n}C_1(a-1)^n + C_2(a-2)^n + \dots + (-1)^n(a-n)^n =$$

[IIT 1972]

- (a) a (b) 0
 (c) a^2 (d) 2^n
 38. If the sum of the coefficients in the expansion of $(\alpha^2 x^2 - 2\alpha x + 1)^{51}$ vanishes, then the value of α is

[IIT 1991; Pb. CET 1988]

- (a) 2 (b) -1
 (c) 1 (d) -2
 39. If $(1+x)^n = C_0 + C_1 x + C_2 x^2 + \dots + C_n x^n$, then the value of $C_0 + 2C_1 + 3C_2 + \dots + (n+1)C_n$ will be

[MP PET 1996; RPET 1997; DCE 1995; AMU 1995; EAMCET 2001; IIT 1971]

- (a) $(n+2)2^{n-1}$ (b) $(n+1)2^n$
 (c) $(n+1)2^{n-1}$ (d) $(n+2)2^n$
 40. The value of $^{15}C_0^2 - ^{15}C_1^2 + ^{15}C_2^2 - \dots - ^{15}C_{15}^2$ is

[MP PET 1996]

- (a) 15 (b) -15
 (c) 0 (d) 51
 41. $2C_0 + \frac{2^2}{2}C_1 + \frac{2^3}{3}C_2 + \dots + \frac{2^{11}}{11}C_{10}$

[MP PET 1999; EAMCET 1992]

- (a) $\frac{3^{11}-1}{11}$ (b) $\frac{2^{11}-1}{11}$
 (c) $\frac{11^3-1}{11}$ (d) $\frac{11^2-1}{11}$

42. If $(1+x)^n = C_0 + C_1 x + C_2 x^2 + \dots + C_n x^n$, then $C_0 C_2 + C_1 C_3 + C_2 C_4 + \dots + C_{n-2} C_n$ equals

[RPET 1996]

- (a) $\frac{(2n)!}{(n+1)!(n+2)!}$ (b) $\frac{(2n)!}{(n-2)!(n+2)!}$
 (c) $\frac{(2n)!}{(n)!(n+2)!}$ (d) $\frac{(2n)!}{(n-1)!(n+2)!}$

43. If $C_0, C_1, C_2, \dots, C_n$ are the binomial coefficients, then $2.C_1 + 2^3.C_3 + 2^5.C_5 + \dots$ equals

[AMU 1999]

- (a) $\frac{3^n + (-1)^n}{2}$ (b) $\frac{3^n - (-1)^n}{2}$
 (c) $\frac{3^n + 1}{2}$ (d) $\frac{3^n - 1}{2}$

44. The sum of coefficients in the expansion of $(x + 2y + 3z)^8$ is

[RPET 2000]

- (a) 3^8 (b) 5^8
 (c) 6^8 (d) None of these

45. $\sum_{k=0}^{10} {}^{20}C_k =$

[Orissa JEE 2004]

- (a) $2^{19} + \frac{1}{2} {}^{20}C_{10}$ (b) 2^{19}
 (c) ${}^{20}C_{10}$ (d) None of these

46. If $S_n = \sum_{r=0}^n \frac{1}{{}^nC_r}$ and $t_n = \sum_{r=0}^n \frac{r}{{}^nC_r}$, then $\frac{t_n}{S_n}$ is equal to

[AIEEE 2004]

- (a) $\frac{2n-1}{2}$ (b) $\frac{1}{2}n-1$
 (c) $n-1$ (d) $\frac{1}{2}n$

47. The coefficient of x in the expansion of $[\sqrt{1+x^2} - x]^{-1}$ in ascending powers of x , when $|x| < 1$, is

[MP PET 1996]

- (a) 0 (b) $\frac{1}{2}$
 (c) $-\frac{1}{2}$ (d) 1

48. If x is positive, the first negative term in the expansion of $(1+x)^{27/5}$ is

[AIEEE 2003]

- (a) 7^{th} term (b) 5^{th} term
 (c) 8^{th} term (d) 6^{th} term

49. Coefficient of x^r in the expansion of $(1-2x)^{-1/2}$ is

[Kurukshetra CEE 2001]

- (a) $\frac{(2r)!}{(r!)^2}$ (b) $\frac{(2r)!}{2^r (r!)^2}$
 (c) $\frac{(2r)!}{(r!)^2 2^{2r}}$ (d) $\frac{(2r)!}{2^r \cdot (r+1)! \cdot (r-1)!}$

50. If x is so small that x^3 and higher powers of x may be neglected, then

$\frac{(1+x)^{\frac{3}{2}} - \left(1 + \frac{1}{2}x\right)^3}{(1-x)^{\frac{1}{2}}}$ may be approximated as

[AIEEE 2005]

- (a) $-\frac{3}{8}x^2$ (b) $\frac{x}{2} - \frac{3}{8}x^2$
 (c) $1 - \frac{3}{8}x^2$ (d) $3x + \frac{3}{8}x^2$

51. If a_1, a_2, a_3, a_4 are the coefficients of any four consecutive terms in the expansion of $(1+x)^n$, then $\frac{a_1}{a_1 + a_2} + \frac{a_3}{a_3 + a_4} =$

[IIT 1975]

- (a) $\frac{a_2}{a_2 + a_3}$ (b) $\frac{1}{2} \frac{a_2}{(a_2 + a_3)}$
 (c) $\frac{2a_2}{a_2 + a_3}$ (d) $\frac{2a_3}{a_2 + a_3}$

52. The number of integral terms in the expansion of $(\sqrt{3} + \sqrt[3]{5})^{256}$ is

[AIEEE 2003]

- (a) 32 (b) 33
 (c) 34 (d) 35

53. For all positive integral values of n , $3^{2n} - 2n + 1$ is divisible by

- (a) 2 (b) 4
 (c) 8 (d) 12

54. If $n \in N$, then $7^{2n} + 2^{3n-3} \cdot 3^{n-1}$ is always divisible by

[IIT 1982]

- (a) 25 (b) 35
 (c) 45 (d) None of these

55. If $n \in N$, then $11^{n+2} + 12^{2n+1}$ is divisible by

[Roorkee 1982]

- (a) 113 (b) 123
 (c) 133 (d) None of these

56. For every positive integer n , $2^n < n!$ when

- (a) $n < 4$ (b) $n \geq 4$
 (c) $n < 3$ (d) None of these

57. Let $P(n)$ denote the statement that $n^2 + n$ is odd. It is seen that $P(n) \Rightarrow P(n+1)$, P_n is true for all

[IIT JEE 1996]

- (a) $n > 1$ (b) n
 (c) $n > 2$ (d) None of these

58. If $P(n) = 2 + 4 + 6 + \dots + 2n$, $n \in N$, then $P(k) = k(k+1) + 2 \Rightarrow P(k+1) = (k+1)(k+2) + 2$ for all $k \in N$. So we can conclude that $P(n) = n(n+1) + 2$ for

- (a) All $n \in N$ (b) $n > 1$
 (c) $n > 2$ (d) Nothing can be said

59. For every natural number n , $n(n+1)$ is always

- (a) Even (b) Odd
 (c) Multiple of 3 (d) Multiple of 4

60. The statement $P(n)$ " $1 \times 1! + 2 \times 2! + 3 \times 3! + \dots + n \times n! = (n+1)! - 1$ " is

- (a) True for all $n > 1$ (b) Not true for any n
 (c) True for all $n \in N$ (d) None of these

61. If the coefficient of x^7 in $\left(ax^2 + \frac{1}{bx}\right)^{11}$ is equal to the coefficient of x^{-7} in $\left(ax - \frac{1}{bx^2}\right)^{11}$, then $ab =$
[MP PET 1999; AMU 2001; Pb. CET 2002; AIEEE 2005]
 (a) 1 (b) $\frac{1}{2}$
 (c) 2 (d) 3
62. The largest term in the expansion of $(3 + 2x)^{50}$ where $x = \frac{1}{5}$ is
[IIT Screening 1993]
 (a) 5th (b) 51st
 (c) 7th (d) 6th
63. The sum of the coefficients in the expansion of $(x + y)^n$ is 4096. The greatest coefficient in the expansion is
[Kurukshetra CEE 1998; AIEEE 2002]
 (a) 1024 (b) 924
 (c) 824 (d) 724
64. For every natural number n , $3^{2n+2} - 8n - 9$ is divisible by
[IIT 1977]
 (a) 16 (b) 128
 (c) 256 (d) None of these
65. If x is positive, the first negative term in the expansion of $(1 + x)^{27/5}$ is
 (a) 7th term (b) 5th term
 (c) 8th term (d) 6th term
66. The number of integral terms in the expansion of $(\sqrt{3} + \sqrt[5]{5})^{256}$ is
 (a) 32 (b) 33
 (c) 34 (d) 35
67. Coefficient of x^2 in the expansion of $\left(x - \frac{1}{2x}\right)^8$ is
 (a) $\frac{1}{7}$ (b) $-\frac{1}{7}$
 (c) -7 (d) 7
68. If the sum of the coefficients in the expansion of $(1 - 3x + 10x^2)^n$ is a and if the sum of the coefficients in the expansion of $(1 + x^2)^n$ is b , then
 (a) $a = 3b$ (b) $a = b^3$
 (c) $b = a^3$ (d) None of these
69. The coefficient of x^5 in the expansion of $(1 + x)^{21} + (1 + x)^{22} + \dots + (1 + x)^{30}$ is
 (a) $^{51}C_5$ (b) 9C_5
 (c) $^{31}C_6 - ^{21}C_6$ (d) $^{30}C_5 + ^{20}C_5$
70. In the binomial expansion of $(a - b)^n$, $n \geq 5$, the sum of the 5th and 6th terms is zero. Then $\frac{a}{b}$ is equal to
 (a) $\frac{1}{6}(n - 5)$ (b) $\frac{1}{5}(n - 4)$
 (c) $\frac{5}{(n - 4)}$ (d) $\frac{6}{(n - 5)}$
71. The number 111.....1 (91 times) is
 (a) Not a prime (b) An even number
 (c) Not an odd number (d) None of these
72. The sum of the series $1 + \frac{1}{5} + \frac{1.3}{5.10} + \frac{1.3.5}{5.10.15} + \dots$ is equal to
 (a) $\frac{1}{\sqrt{5}}$ (b) $\frac{1}{\sqrt{2}}$
 (c) $\sqrt{\frac{5}{3}}$ (d) $\sqrt{5}$
73. $\binom{n}{0} + 2\binom{n}{1} + 2^2\binom{n}{2} + \dots + 2^n\binom{n}{n}$ is equal to
 (a) 2^n (b) 0
 (c) 3^n (d) None of these
74. The sum of the coefficients in the expansion of $(x + y)^n$ is 4096. The greatest coefficient in the expansion is
 (a) 1024 (b) 924
 (c) 824 (d) 724
75. The greatest integer which divides the number $101^{100} - 1$, is
 (a) 100 (b) 1000
 (c) 10000 (d) 100000
76. If $(1 + x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$, then the value of $C_0 + 2C_1 + 3C_2 + \dots + (n + 1)C_n$ will be
 (a) $(n + 2)2^{n-1}$ (b) $(n + 1)2^n$
 (c) $(n + 1)2^{n-1}$ (d) $(n + 2)2^n$
77. If $\frac{T_2}{T_3}$ in the expansion of $(a + b)^n$ and $\frac{T_3}{T_4}$ in the expansion of $(a + b)^{n+3}$ are equal, then $n =$
 (a) 3 (b) 4
 (c) 5 (d) 6
78. If $(1 - x + x^2)^n = a_0 + a_1x + a_2x^2 + \dots + a_{2n}x^{2n}$, then $a_0 + a_2 + a_4 + \dots + a_{2n} =$
 (a) $\frac{3^n + 1}{2}$ (b) $\frac{3^n - 1}{2}$
 (c) $\frac{1 - 3^n}{2}$ (d) $3^n + \frac{1}{2}$
79. In the expansion of $(1 + x)^n$ the sum of coefficients of odd powers of x is
 (a) $2^n + 1$ (b) $2^n - 1$
 (c) 2^n (d) 2^{n-1}
80. If for positive integers $r > 1$, $n > 2$, the coefficient of the $(3r)^{th}$ and $(r + 2)^{th}$ powers of x in the expansion of $(1 + x)^{2n}$ are equal, then
 (a) $n = 2r$ (b) $n = 3r$
 (c) $n = 2r + 1$ (d) None of these
81. If the coefficients of r^{th} term and $(r + 4)^{th}$ term are equal in the expansion of $(1 + x)^{20}$, then the value of r will be
 (a) 7 (b) 8
 (c) 9 (d) 10
82. The total number of terms in the expansion of $(x + a)^{100} + (x - a)^{100}$ after simplification will be
 (a) 202 (b) 51
 (c) 50 (d) None of these

Answer Sheet									
1	2	3	4	5	6	7	8	9	10
c	d	c	d	c	a	c	c	d	b
11	12	13	14	15	16	17	18	19	20
d	a	b	b	c	c	d	c	b	b
21	22	23	24	25	26	27	28	29	30
c	b	a	b	b	b	a	c	b	c
31	32	33	34	35	36	37	38	39	40
c	b	b	a	c	a	b	c	a	c
41	42	43	44	45	46	47	48	49	50
a	b	b	c	a	d	d	c	b	a
51	52	53	54	55	56	57	58	59	60
c	b	a	a	c	b	d	d	a	c
61	62	63	64	65	66	67	68	69	61
a	c	b	a	c	b	c	b	c	b
71	72	73	74	75	76	77	78	79	80
a	c	c	b	c	a	c	a	d	c
81	82								
c	b								